Port

USB-C source, ESD, lead

File: port.kicad_sch

DATA port v1.0 — USB-C source (Rp 56k on CC1/CC2), USBLC6 ESD, 8-pin JST PH lead to the bay node's J7.
Lead pinout here: 1 INT, 2 SCL, 3 SDA (all three open), 4 GND, 5 D−, 6 D+, 7 GND, 8 VBUS
  (the node's J7 mirrored: one 'Same direction' PH 8 cable).

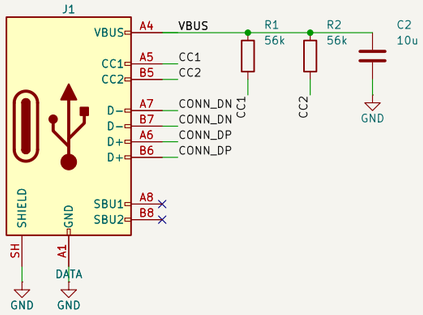
Generated from electronics/data-port/design.py — edit the Python, then regenerate
The drawer's USB-C port for the board under test, wired to the bay node (hw-testing-rack)
hw-testing-rack

Sheet: /
File: data-port.kicad_sch

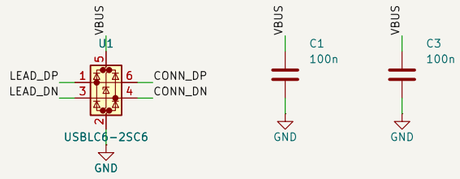
Title: DATA port — Overview

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/2

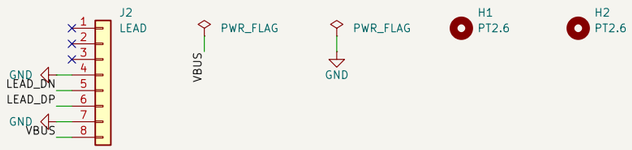
USB-C receptacle (source): Rp 56k on each CC, the shell to GND, 10 µF on VBUS



ESD on the pair, at the receptacle



Lead to the bay node's DATA header (J7): 4 GND, 5 D−, 6 D+, 7 GND, 8 VBUS; 1-3 (INT, I²C) open



Test pads (bottom side, labelled): no short to GND, then VBUS at power-up



Rp 56k to VBUS on CC1 and CC2: 'default USB power', as on the DATA ×4; not 22k (1.5 A): the node's TPS2553D switch (DUT_5V) limits at 1.34-1.52 A. The node turns the board under test on and off.
The lead's pinout is the node's J7 mirrored: a 'Same direction' PH 8 cable (pin n to pin 9−n) joins them; pins 1-3 (the node's INT and I²C) stay open here.

Generated from electronics/data-port/design.py — edit the Python, then regenerate
The drawer's USB-C port for the board under test, wired to the bay node (hw-testing-rack)
hw-testing-rack

Sheet: /Port/		
File: port.kicad_sch		
Title: DATA port — USB-C source, ESD, lead		
Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/2